

Shakir Showkat Sofi

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🔗 Personal website

Doctoral Researcher in Electrical Engineering (ESAT) at KU Leuven, Belgium, under the supervision of Prof. Lieven De Lathauwer. Research focuses on tensor networks and tensor decomposition methods for data compression and completion, with applications in machine learning, quantum tomography, and spatiotemporal forecasting.

EDUCATION

February, 2023 Present	KU Leuven, ESAT, STADIUS, KULAK KORTRIJK, BE Ph.D. Engineering Science (Electrical Engineering) Advisor : Prof. Lieven De Lathauwer (ESAT, STADIUS and Group Science, E & T) > Topic : “Learning tensor networks from incomplete data.” Tensor decompositions, Tensor networks, Data compression and completion, Quantum state tomography.
September, 2020 June, 2022	Skolkovo Institute of Science and Technology, Moscow, RU M.Sc. Data Science GPA : 4.7/5 Advisor : Prof. Ivan Oseledets (Skoltech & RAS) > Thesis : “Spatiotemporal forecasting with application to the weather forecast” Matrix and tensor completion, Spatiotemporal forecasting, Machine learning
August, 2015 August, 2019	Islamic University Of Science And Technology, SRINAGAR, J&K, IN B.Tech. Electric Engineering GPA : 9.18/10 Advisor : Dr. Shahkar A. Nahvi (IUST & IIT-Delhi) > Thesis : “Energy-Based Modeling of Dc-Dc Power Converters” Dc-Dc power converters, Euler-Lagrangian modelling, Simulation

EXPERIENCE

2024–Present	Flanders AI Research Program, FLANDERS, Belgium Research Project > Developing tensor-based methods for sports analytics.
2025–Present	Tensor Decompositions and Decision Trees, KU LEUVEN, Belgium Research Project > Investigating the integration of tensor decomposition methods with decision-tree learning.
2023- 2025	pyTensorlab, KU LEUVEN, Belgium Software Development Contribution > Contributed to the development of the Python package for tensor computations, including demo preparation and contributions to tensorization and visualization functionalities.
June 2026	EECI International Graduate School on Control, KU LEUVEN, Belgium Graduate Summer School > Deep Learning for System Identification.
Summer 2021	Skolkovo Institute of Science and Technology & TENSOR FIELD, Moscow, Russia Research Internship > Data-driven weather forecasting and spatiotemporal time-series analysis.

Summer 2021 | Global Quantum Network (QWorld), VIRTUAL,
Summer School
‣ Quantum computing and programming with Qiskit.

PUBLICATIONS

For the full list of publications, please check my [Google Scholar page](#).

TALKS

For an up-to-date list of invited and conference talks, please visit my [personal webpage](#).

SKILLS

</> **Programming and Writing :** C, Python, PyTorch, Matlab, V.B, HTML, SQL, $\text{\LaTeX}$, LyX, Ms Office.
 **Languages :** English, Urdu, Kashmiri.

MEMBERSHIPS

- Society for Industrial and Applied Mathematics (SIAM)
- European Association for Signal Processing (EURASIP)

TEACHING AND MENTORING

T.A. & Lecturer	KU Leuven, CAMPUS KULAK KORTRIJK, Belgium ‣ Applied AI : Tensors & AI Delivered lectures on tensor decompositions and their applications in machine learning and data-driven modeling. ‣ Systems and Control Theory Conducted exercise sessions, developed theoretical and programming assignments, and assisted with assessment and grading.
Lecturer	KIMS – Quantum AI Lab, Srinagar, J & K, India ‣ Python Programming : Foundations and Applications Delivered a module focusing on linear algebra and its applications.

CERTIFICATIONS AND SCHOLARSHIPS

2015-2019 Merit-Cum Means Scholarship during Undergrad. by MHRD, Govt. of India.
2020 Graduate fellowship at Skolkovo Institute of Science and Technology, Moscow, RU
2023 Doctoral scholarships at KU Leuven, Belgium.

REFERENCES

Available upon request.